Dr. Joshua D. Lothringer

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Research Interests

Observing, modeling, and retrieving planetary, exoplanetary, brown dwarf, and stellar atmospheres.

Positions

Assistant Astronomer
Space Telescope Science Institute, Baltimore, MD — 01/2024 – Present

Assistant Research Scientist
William H. Miller III Dept. of Physics and Astronomy, Johns Hopkins University, Baltimore, MD — 01/2025 – Present

Assistant Professor
Department of Physics, Utah Valley University, Orem, UT — 08/2021 – 12/2023

Postdoctoral Fellow
Department of Physics and Astronomy, Johns Hopkins University, Baltimore, MD — 08/2019 – 08/2021

Graduate Assistant / Associate
Lunar and Planetary Laboratory, University of Arizona, Tucson, AZ — 08/2014 – 08/2019

Command Controller
Mission Operations and Data Systems, Laboratory for Atmospheric and Space Physics, University of Colorado at Boulder, Boulder, CO — 05/2012 – 08/2014

Education

Doctor of Philosophy, Planetary Science
University of Arizona, Tucson, AZ
Advisor: Prof. Travis Barman
Dissertation: Characterizing the Atmospheres of Planet Populations: From Sub-Jovian to Ultra-hot Jupiter Exoplanets — 08/2014 – 08/2019

Master of Science, Planetary Science
University of Arizona, Tucson, AZ — 12/2016

Bachelor of Arts, Astronomy (Minor: Philosophy)
University of Colorado, Boulder, CO — 08/2010 – 12/2013

Publications *(including submitted)*

First-Author

1. **Lothringer, J. D.**, Lowson, N., Fu, G. “The Library of Exoplanet Atmospheric Composition Measurements: Population Level Trends in Exoplanet Composition with ExoComp", 2026, AJ,

171, 31.

2. **Lothringer, J. D.**, Bennett, K. A., Sing, D. K., et al., "Refractory and Volatile Species in the UV-to-IR Transmission Spectrum of Ultra-hot Jupiter WASP-178b with HST and JWST", 2025, AJ, 169, 274.
3. **Lothringer, J. D.**, Zhou, Y., Apai, D., et al., "Atmospheric Retrievals of the Phase-resolved Spectra of Irradiated Brown Dwarfs WD-0137B and EPIC-2122B", 2024, ApJ, 968, 126.
4. **Lothringer, J.D.**, Sing, D. K., Rustamkulov, Z., et al. "UV absorption by silicate cloud precursors in ultra-hot Jupiter WASP-178b", 2022, *Nature*, 604, 49.
5. **Lothringer, J.D.**, Rustamkulov, Z., Sing, D. K., et al. "A New Window into Planet Formation and Migration: Refractory-to-Volatile Elemental Ratios in Ultra-hot Jupiters", 2021, ApJ, 914, 12.
6. **Lothringer, J.D.**, & Casewell, S. L. "Atmosphere Models of Brown Dwarfs Irradiated by White Dwarfs: Analogs for Hot and Ultrahot Jupiters", 2020, ApJ, 905, 163.
7. **Lothringer, J.D.**, Fu, G., Sing, D. K., & Barman, T. S. "UV Exoplanet Transmission Spectral Features as Probes of Metals and Rainout", 2020, ApJL, 898, L14.
8. **Lothringer, J.D.**, & Barman, T. S. "The PHOENIX Exoplanet Retrieval Algorithm and Using H^- Opacity as a Probe in Ultrahot Jupiters", 2020, AJ, 159, 289.
9. **Lothringer, J.D.**, & Barman, T. "The Influence of Host Star Spectral Type on Ultra-hot Jupiter Atmospheres", 2019, ApJ, 876, 69.
10. **Lothringer, J.D.**, Barman, T., & Koskinen, T. "Extremely Irradiated Hot Jupiters: Non-oxide Inversions, H^- Opacity, and Thermal Dissociation of Molecules", 2018, ApJ, 866, 27.
11. **Lothringer, J.D.**, Benneke, B., Crossfield, I. J. M., et al. "An HST/STIS Optical Transmission Spectrum of Warm Neptune GJ 436b", 2018, AJ, 155, 66.

Co-Author

12. *Feinstein, A. D., Booth, R. A., Bergner, J. B.,* ***Lothringer, J.D.****, et al., "On Linking Planet Formation Models, Protoplanetary Disk Properties, and Mature Gas Giant Exoplanet Atmospheres", 2025, Submitted (PASP), arXiv:2506.00669.*
13. Kanumalla, K., Line, M. R., Chiarella, M., et al., "The Roasting Marshmallows Program with IGRINS on Gemini South V: Atmosphere of MASCARA-1b is Enriched in Refractory Elements", 2026, Accepted (AJ), arXiv:2606.07497.
14. Mukherjee, S., Sing, D.K., Fu, G., et al., "Cloudy Mornings and clear evenings on a gas giant exoplanet", 2026, *Science*, 392, 858-862.
15. Ashtari, R., Collins, S., Splinter, J., et al., "Heat Reveals What Clouds Conceal: Global Carbon and Longitudinally Asymmetric Chemistry on LTT 9779 b", 2026, AJ, 171, 215.
16. Wang, L.-C., Rustamkulov, Z., Sing, D. K., et al., "A Comprehensive Analysis of the Panchromatic Transmission Spectrum of the Hot-Saturn WASP-96b", 2026, AJ, 171, 147.
17. Baldwin, A., **Lothringer, J. D.**, Santos, L. A. dos ., et al., "A High-resolution Near-Ultraviolet Transmission Spectrum of KELT-9b: Mg II and Fe II Escaping from the Hottest Known Giant Planet", 2026, AJ, 171, 122.
18. Crossfield, I. J. M., Ahrer, E.-M., Brande, J., et al., "Mapping the SO_2 Shoreline in Gas Giant Exoplanets", 2025, ApJ, 994, 184.
19. Chachan, Y., **Lothringer, J.D.**, Inglis, J., et al., "Strong NUV Refractory Absorption and Dissociated Water in the Hubble Transmission Spectrum of KELT-20b", 2025, AJ, 170, 234.
20. Bennett, K. A., MacDonald, R. J., Peacock, S., et al., "Additional JWST/NIRSpec Transits of

GJ 1132b Reveal a Featureless Spectrum", 2025, AJ, 170, 205.

21. Fu, G., Mukherjee, S., Stevenson, K. B., et al., "Overcast Mornings and Clear Evenings in Hot Jupiter Exoplanet Atmospheres", 2025, ApJL, 989, L17.
22. Evans-Soma, T. M., Sing, D. K., Barstow, J. K., et al., "SiO and a super-stellar C/O ratio in the atmosphere of WASP-121b", 2025, Nature Astronomy, 9, 845.
23. Fu, G., Stevenson, K. B., Sing, D. K., et al., "Statistical Trends in JWST Transiting Exoplanet Atmospheres", 2025, ApJ, 986, 1.
24. Gapp, C., Evans-Soma, T. M., Barstow, J. K., et al., "WASP-121b's Transmission Spectrum Observed with JWST/NIRSpec G395H", 2025, AJ, 169, 341.
25. Balmer, W. O., Kammerer, J., Pueyo, L., et al., "JWST-TST High Contrast: NIRCam Bar Coronagraphy Reveals CO_2 in HR 8799 and 51 Eri Exoplanets' Atmospheres", 2025, AJ, 169, 209.
26. Amaro, R. C., Apai, D., Zhou, Y., et al., "Phase-resolved HST WFC3 Spectroscopy of White Dwarf–Brown Dwarf Binaries and Energy Redistribution–Irradiation Trends", 2025, ApJ, 979, 231.
27. Bennett, K. A., Sing, D. K., Stevenson, K. B., et al., "An HST Transmission Spectrum of LTT 1445Ab", 2025, AJ, 169, 111.
28. Pelletier, S., Benneke, B., Chachan, Y., et al., "CRIRES$^+$ and ESPRESSO Reveal an Atmosphere Enriched in Volatiles on WASP-121b", 2025, AJ, 169, 10.
29. Sing, D. K., Evans-Soma, T. M., Rustamkulov, Z., et al., "An Absolute Mass, Precise Age, and Hints of Planetary Winds for WASP-121A and b from JWST NIRSpec", 2024, AJ, 168, 231.
30. Iskandarli, L., Farihi, J., **Lothringer, J. D.**, et al., "Novel constraints on companions to the Helix nebula central star", 2024, MNRAS, 534, 3498.
31. French, J. R., Casewell, S. L., Amaro, R. C., et al., "The only inflated brown dwarf in an eclipsing white dwarf–brown dwarf binary: WD1032+011B", 2024, MNRAS, 534, 2244.
32. Fu, G., Welbanks, L., Deming, D., et al., "Hydrogen sulfide and metal-enriched atmosphere for a Jupiter-mass exoplanet", 2024, *Nature*, 632, 752.
33. Carter, A. L., May, E. M., Espinoza, N., et al., "A benchmark JWST near-infrared spectrum for WASP-39b", 2024, Nature Astronomy, 8, 1008.
34. Bell, T. J., Crouzet, N., Cubillos, P. E., et al., "Nightside clouds and disequilibrium chemistry on WASP-43b", 2024, Nature Astronomy, 8, 879.
35. Amaro, R. C., Apai, D., Lew, B. W. P., et al., "Inefficient Day-to-night Heat Redistribution in Irradiated Brown Dwarf SDSS 1557B", 2024, ApJ, 966, 4.
36. Kirk, J., Stevenson, K. B., Fu, G., et al., "JWST/NIRCam Transmission Spectroscopy of the Sub-Earth GJ 341b", 2024, AJ, 167, 90.
37. Powell, D., Feinstein, A. D., Lee, E. K. H., et al., "Sulfur dioxide in the mid-infrared transmission spectrum of WASP-39b", 2024, *Nature*, 626, 979.
38. Brande, J., Crossfield, I. J. M., Kreidberg, L., et al., "Clouds and Clarity: Revisiting Atmospheric Feature Trends in Neptune-size Exoplanets", 2024, ApJL, 961, L23.
39. Reggiani, H., Galarza, J. Y., Schlaufman, K. C., et al., "Composition of the Parent Protoplanetary Disk of β Pic b Revealed by the β Pic Moving Group Member HD 181327", 2024, AJ, 167, 45.
40. May, E. M., MacDonald, R. J., Bennett, K. A., et al., "Double Trouble: Two Transits of GJ 1132b with JWST NIRSpec G395H", 2023, ApJL, 959, L9.
41. Lustig-Yaeger, J., Fu, G., May, E. M., et al., "A JWST transmission spectrum of the nearby

Earth-sized exoplanet LHS 475b", 2023, Nature Astronomy, 7, 1317.

42. Roy, P.-A., Benneke, B., Piaulet, C., et al., “Water Absorption in the Transmission Spectrum of GJ 9827d", 2023, ApJL, 954, L52.
43. Coria, D. R., Crossfield, I. J. M., **Lothringer, J.D.**, et al. “The Missing Link: Multi-isotopic Abundances in Solar Twin Stars", 2023, ApJ, 954, 121.
44. Coulombe, L.-P., Benneke, B., Challener, R., et al. “A broadband thermal emission spectrum of ultra-hot Jupiter WASP-18b", 2023, *Nature*, 620, 292.
45. van Sluijs, L., Birkby, J. L., **Lothringer, J.D.**, et al., “Carbon monoxide emission lines reveal an inverted atmosphere in WASP-33b", 2023, MNRAS, 522, 2145.
46. Tsai, S.-M., Lee, E. K. H., Powell, D., et al., “Photochemically produced SO_2 in the atmosphere of WASP-39b", 2023, *Nature*, 617, 483.
47. Grant, D., **Lothringer, J.D.**, Wakeford, H. R., et al., “Detection of Carbon Monoxide’s 4.6 Micron Fundamental Band Structure in WASP-39b with JWST NIRSpec G395H", 2023, ApJL, 949, L15.
48. Moran, S. E., Stevenson, K. B., Sing, D. K., et al., “Water-rich Atmosphere or Stellar Contamination for GJ 486b from JWST", 2023, ApJL, 948, L11.
49. Amaro, R. C., Apai, D., Zhou, Y., et al., “Hotter than Expected: HST/WFC3 Phase-resolved Spectroscopy of a Rare Irradiated Brown Dwarf", 2023, ApJ, 948, 129.
50. Gressier, A., Lecavelier des Etangs, A., Sing, D. K., et al., “The Hubble PanCET program: NUV transmission spectrum of WASP-79b", 2023, A&A, 672, A34.
51. Feinstein, A. D., Radica, M., Welbanks, L., (**Lothringer, J.D.** 1st tier) et al., “Early Release Science of WASP-39b with JWST NIRISS", 2023, *Nature*, 614, 670.
52. Alderson, L., Wakeford, H. R., Alam, M. K., (**Lothringer, J.D.** 1st tier) et al., “Early Release Science of WASP-39b with JWST NIRSpec G395H", 2023, *Nature*, 614, 664.
53. Rustamkulov, Z., Sing, D. K., Mukherjee, S., (**Lothringer, J.D.** 1st tier) et al., “Early Release Science of WASP-39b with JWST NIRSpec PRISM", 2023, *Nature*, 614, 659.
54. Ahrer, E.-M., Stevenson, K. B., Mansfield, M., et al., “Early Release Science of WASP-39b with JWST NIRCam", 2023, *Nature*, 614, 653.
55. JWST Transiting Exoplanet Community ERS Team, Ahrer, E.-M., Alderson, L., (**Lothringer, J.D.** 1st tier) et al., “Identification of carbon dioxide in an exoplanet atmosphere", 2023, *Nature*, 614, 649.
56. Mikal-Evans, T., Sing, D. K., Dong, J., et al. “A JWST NIRSpec Phase Curve for WASP-121b", 2023, ApJL, 943, L17.
57. Chachan, Y., Knutson, H. A., **Lothringer, J.D.**, & Blake, G. A. “Breaking Degeneracies in Formation Histories by Measuring Refractory Content in Gas Giants", 2023, ApJ, 943, 112.
58. Kasper, D., Bean, J. L., Line, M. R., et al., “Unifying High- and Low-resolution Observations to Constrain the Dayside Atmosphere of KELT-20b/MASCARA-2b", 2023, AJ, 165, 7.
59. Fu, G., Espinoza, N., Sing, D. K., et al., “Water and an Escaping Helium Tail in HAT-P-18b from JWST NIRISS/SOSS", 2022, ApJL, 940, L35.
60. Brande, J., Crossfield, I. J. M., Kreidberg, L., et al., “A Mirage or an Oasis? Water Vapor in the Atmosphere of the Warm Neptune TOI-674b", 2022, AJ, 164, 197.
61. Kreidberg, L., Molière, P., Crossfield, I. J. M., et al., “Tentative Evidence for Water Vapor in HD 106315c", 2022, AJ, 164, 124.
62. Buzard, C., Casewell, S. L., **Lothringer, J.D.**, & Blake, G. A., “Near-infrared Spectra of the

Inflated Brown Dwarf NLTT 5306B", 2022, AJ, 163, 262.

63. Gibson, N. P., Nugroho, S. K., **Lothringer, J.D.**, Maguire, C., & Sing, D. K., “Relative abundance constraints from high-resolution transmission spectroscopy of WASP-121b", 2022, MNRAS, 512, 4618.

64. Fu, G., Sing, D. K., Deming, D., et al., “The Hubble PanCET Program: Emission Spectrum of HAT-P-41b", 2022, AJ, 163, 190.

65. Reggiani, H., Schlaufman, K. C., Healy, B. F., **Lothringer, J.D.**, & Sing, D. K., “Evidence that WASP-77Ab Formed Beyond its Protoplanetary Disk’s H_2O Ice Line", 2022, AJ, 163, 159.

66. Bruno, G., Lewis, N. K., Valenti, J. A., et al., “Hiding in plain sight: observing planet-starspot crossings with JWST", 2022, MNRAS, 509, 5030.

67. Fu, G., Sing, D. K., **Lothringer, J.D.**, et al., “Strong H_2O and CO Emission Features in the Spectrum of KELT-20b Driven by Stellar UV Irradiation", 2022, ApJL, 925, L3.

68. Zhou, Y., Apai, D., Tan, X., et al., “HST/WFC3 Complete Phase-resolved Spectroscopy of WD 0137 and EPIC 2122", 2022, AJ, 163, 17.

69. Fu, G., Deming, D., May, E., et al., “The Hubble PanCET program: Transit and Eclipse Spectroscopy of WASP-74b", 2021, AJ, 162, 271.

70. Sainsbury-Martinez, F., Casewell, S. L., **Lothringer, J.D.**, Phillips, M. W., & Tremblin, P., “Exploring deep and hot adiabats as a potential solution to radius inflation in brown dwarfs", 2021, A&A, 656, A128.

71. Merritt, S. R., Gibson, N. P., Nugroho, S. K., et al., “An inventory of atomic species in the atmosphere of WASP-121b", 2021, MNRAS, 506, 3853.

72. Fu, G., Deming, D., **Lothringer, J.D.**, et al., “The Hubble PanCET Program: Transit and Eclipse Spectroscopy of WASP-76b", 2021, AJ, 162, 108.

73. Wilson, J., Gibson, N. P., **Lothringer, J.D.**, et al., “Gemini/GMOS optical transmission spectroscopy of WASP-121b", 2021, MNRAS, 503, 4787.

74. Mikal-Evans, T., Crossfield, I. J. M., Benneke, B., et al., “Transmission Spectroscopy for HD 3167c: Evidence for Molecular Absorption", 2021, AJ, 161, 18.

75. Guo, X., Crossfield, I. J. M., Dragomir, D., et al., “Updated Parameters and a New Transmission Spectrum of HD 97658b", 2020, AJ, 159, 239.

76. Gibson, N. P., Merritt, S., Nugroho, S. K., et al., “Detection of Fe I in WASP-121b and a new likelihood-based approach for Doppler-resolved spectroscopy", 2020, MNRAS, 493, 2215.

77. Turner, J. D., de Mooij, E. J. W., Jayawardhana, R., et al., “Detection of Ionized Calcium in the Atmosphere of KELT-9b", 2020, ApJL, 888, L13.

78. Benneke, B., Wong, I., Piaulet, C., et al., “Water Vapor and Clouds on Habitable-zone Sub-Neptune K2-18b", 2019, ApJL, 887, L14.

79. Benneke, B., Knutson, H. A., **Lothringer, J.D.**, et al., “A sub-Neptune exoplanet with a low-metallicity methane-depleted atmosphere and Mie-scattering clouds", 2019, Nature Astronomy, 3, 813.

80. Steinrueck, M. E., Parmentier, V., Showman, A. P., **Lothringer, J.D.**, & Lupu, R. E., “Disequilibrium Carbon Chemistry on HD 189733b: Atmospheric Structure and Phase Curves", 2019, ApJ, 880, 14.

81. Crossfield, I. J. M., **Lothringer, J.D.**, Flores, B., et al., “Unusual Isotopic Abundances in a Fully Convective Stellar Binary", 2019, ApJL, 871, L3.

82. Fossati, L., Koskinen, T., **Lothringer, J.D.**, et al., “Extreme-ultraviolet Radiation from A-stars: Implications for Ultra-hot Jupiters", 2018, ApJL, 868, L30.

83. Bean, J. L., Stevenson, K. B., Batalha, N. M., et al., "The Transiting Exoplanet Community Early Release Science Program for JWST", 2018, PASP, 130, 114402.
84. Kilpatrick, B. M., Cubillos, P. E., Stevenson, K. B., et al., "Community Targets of JWST's ERS Program: Evaluation of WASP-63b", 2018, AJ, 156, 103.
85. Bell, T. J., Nikolov, N., Cowan, N. B., et al., "The Very Low Albedo of WASP-12b from Spectral Eclipse Observations with Hubble", 2017, ApJL, 847, L2.
86. Stevenson, K. B., Lewis, N. K., Bean, J. L., et al., "Transiting Exoplanet Studies and Community Targets for JWST's ERS Program", 2016, PASP, 128, 094401.
87. Crossfield, I. J. M., Ciardi, D. R., Petigura, E. A., et al., "197 Candidates and 104 Validated Planets in K2's First Five Fields", 2016, ApJS, 226, 7.

Proceedings and Other Publications

1. **Lothringer, J.D.**, Carlberg, J.K., Lockwood, S. "Barycentric Corrections for HST/STIS Data", 2026, HST STIS Instrument Science Report 2026-02. arXiv:2606.12244
2. **Lothringer, J.D.**, et al. "Characterizing Transiting Exoplanet Atmospheres in the 2030s with the Hubble Space Telescope", 2026, White Paper, "Building a Roadmap for *Hubble* science into the 2030s". arXiv e-prints.
3. Alam, M., et al. "HST Notebooks First Release", 2025, Zenodo. Version v1.0.0. 10.5281/zenodo.17179608
4. **Lothringer, J. D.** et al. "Uncertainties in Low-Count STIS Spectra", 2025, HST STIS Instrument Science Report 2025-04. arXiv:2601.18910.
5. Ardila, D. R., et al. "The UV-SCOPE Mission: Ultraviolet Spectroscopic Characterization Of Planets and their Environments", 2022, Proceedings of the SPIE. id. 1218104. arXiv:2208.09547.
6. **Lothringer, J. D.** "Stellar specific intensity models used in 'Hiding in plain sight: observing planet-starspot crossings with the James Webb Space Telescope'", 2021, Zenodo Software package, id. 5609421.

Select Invited Talks

1. Caltech/IPAC Lunch Seminar, *Virtual.* Jun. 2026.
2. Astronomy Colloquium, University of Maryland, College Park, MD. Apr. 2026.
3. "JWST Exoplanet Observations", Space Sciences Board, National Academies of Sciences, Engineering, and Medicine, Washington D. C., Jul. 2025.
4. Exocoffee Seminar on Lothringer et al. 2025, Max Planck Institute for Astronomy, Heidelberg, Germany. *Virtual.* May 2025.
5. The Great Link Workshop, MPIA, Heidelberg, Germany. Jul. 2024.
6. University Town Hall with President Astrid Tuminez, Utah Valley University, Orem, UT. Sep. 2023.
7. Convocation Lecture Series & Science Division Seminar, Snow College, Ephriam, UT. Aug. 2023.
8. Astronomy Colloquium, Anton Pannekoek Institute for Astronomy, University of Amsterdam. Nov. 2022.
9. Physics and Astronomy Colloquium, Brigham Young University, Provo, UT. Oct. 2022.
10. Brown Dwarf to Exoplanet Connection in the Era of JWST Splinter Session, Exoplanets IV, Las Vegas. May 2022.

11. Physics Colloquium, Department of Physics, Utah Valley University, Orem, UT. Sep. 2021.
12. HotSci, Space Telescope Science Institute. Virtual. Jul. 2021.
13. Exoplanet Lunch, Center for Astrophysics, Harvard University. Virtual. Jan. 2021.
14. Exoplanet Journal Club, Jet Propulsion Laboratory. Virtual. Jan. 2021.
15. Star and Planet Seminar, Imperial College London. Virtual. Oct. 2020.
16. Exocoffee, Max Planck Institute for Astronomy. Virtual. May 2020.
17. Exoplanet Tea, Massachusetts Institute of Technology, Cambridge, MA. Oct. 2019.
18. Exoplanet Lunch, Center for Astrophysics, Harvard University, Cambridge, MA. Oct. 2019.
19. Wine & Cheese Seminar, Center for Astrophysical Sciences, Johns Hopkins University, Baltimore, MD. Sep. 2019.
20. Theoretical Astrophysics Program Graduate Research Prize Talk, University of Arizona, Tucson, AZ. Apr. 2019.
21. Exoplanet Seminar, DTU Space, Lyngby, Denmark. Feb. 2019.
22. Star and Planet Formation Seminar, Max Planck Institute for Astronomy, Heidelberg, Germany. Jul. 2016.

Select Contributed Conference Presentations

1. "The Same But Different: SiO2 Clouds in Two L-dwarfs with Varying Properties from JWST." The Brown Dwarf to Exoplanet Connection 4. University of Delaware, Newark, DE. Apr. 2026. *Oral.*
2. "The UVOIR Transit Spectrum of Ultra-hot Jupiter WASP-178b with HST and JWST." 245th AAS Winter Meeting. National Harbor, MD. Jan. 2025. *Oral.*
3. "Atmospheric Retrievals of the Phase-resolved Spectra of Two Hot and Ultra-hot Jupiter Analogs." 241st AAS Winter Meeting. Seattle, WA. Jan. 2023. *Oral.*
4. "The UV Transmission Spectrum of Ultra-hot Jupiter WASP-178b." Exoplanets IV, Las Vegas. May 2022. *Oral.*
5. "The Importance of UV Opacity in Extremely Irradiated Objects." Stars and Planets in the UV. Virtual. May 2021. *Oral.*
6. "Re-Interpreting UV-Optical Transmission Spectra of Hot and Ultra-Hot Jupiters." 237th AAS Winter Meeting. Virtual. Jan. 2021. *Oral.*
7. "Understanding Ultra-hot Jupiters Through Irradiated Brown Dwarfs." 235th AAS Winter Meeting, Honolulu, HI. Jan. 2020. *Oral.*
8. "Highly Irradiated Brown Dwarfs as High-mass Ultra-hot Jupiters." BDExoCon, University of Delaware, Newark, DE. Oct. 2019. *Oral.*
9. "Characterizing the Atmospheres of Exoplanet Populations: From Sub-Jovian to Ultra-hot Jupiter Exoplanets." AAS Winter Meeting, Seattle, WA. Jan. 2019. *Oral.*
10. "Modeling the Most Extreme Jovian Atmospheres." Exoplanets Around Hot Stars, Vanderbilt University, Nashville, TN. Jun. 2018. *Oral.*
11. "Self-Consistent Atmosphere Models of the Most Extreme Hot Jupiters." AAS Winter Meeting, Washington D.C. Jan. 2018. *Oral.*

Honors, Awards, and Grants *(+$2,300,000 total)*

- **PI, NSF S-STEM Program** (*2022–2023*)

  - “Pro-STEM: Promoting Engagement in Chemistry, Physics, and Earth Sciences"
  - Six-year, $1.5M scholarship, mentorship, and research program supporting 25+ students.
- **PI/Co-PI, *James Webb Space Telescope* Programs**
  - Program 2055 (9.1 hrs; $100,211) — “Tracing Hot Jupiter Formation and Migration with Volatile and Refractory Element Ratios"
  - Program 2288 (7.4 hrs; $113,998; Co-PI: Jeff Valenti) — “Formation and Impact of Silicate Clouds on L Dwarfs"
  - Program 7255 (52.3 hrs; $308,468; PI: Kevin Stevenson) — “A Day In The Life of KELT-9b"
  - Program 10290 (9.9 hrs; $128,024) — “Mistaken Identity? Resolving Ultra-hot Jupiter NUV Absorption"
  - Program 10674 (117.5 hrs; $635,767) — “Comparative Planetology in Kepler-89: Mass-Loss, Disk Gaps, and Super-Puffs"
- **PI, *Hubble Space Telescope* Programs**
  - Program 16086 ($86,995; 10 orbits) — “Comparing Escaping Metals and Heat Deposition in Ultra-hot Jupiters"
  - Program 16142 ($99,319; AR Theory) — “The First Grid of White-Dwarf-Irradiated Brown Dwarf Atmosphere Models"
  - Program 16270 ($63,530; 20 orbits) — “Heavy Metal Bands: A Study of Escaping Ions from the Hottest Jovian Atmospheres"
  - Program 16450 ($45,465; 10 orbits) — “Measuring the Rock-to-Ice Ratio in an Exoplanet"
  - PI of 3 additional Calibration HST Programs
- Co-I on 12 *JWST* Programs (500+ hours); 21 *HST* Programs (300+ orbits); 1 *Spitzer* Program (61 hours); 1 NASA XRP Program (PI: Schlaufman; 2022–2025); 1 *Subaru* Intensive Program (31 nights)
- Director’s Discretionary Research Funds: CHEXO Funding ($3,400) 2024
- College of Science Dean’s Award of Excellence for Scholarship ($2,750) 2023
- Scholarly Activities Committee Dissemination Grant ($1,344) 2022
- Theoretical Astrophysics Program Graduate Research Prize ($500) 2019
- Galileo Circle Scholar ($2,000) 2019
- Galileo Circle Scholar ($1,000) 2016
- 1st Place — The Art of Planetary Science, Data Art Category 2015
- Graduate and Professional Student Council Travel Grant ($250) 2015
- 2015 Sagan Workshop Travel Grant ($700) 2015
- Science Phoenix Award — SORCE Mission Operations 2014

Technical Experience

HST/STIS Instrument Team
Pipeline Block Lead; Contact Scientist; ETC Working Group; CCD Working Group; PI/co-I on 5 calibration programs 2024 – 2026

JWST Science Policy Division 2024 – Present
JWST TAC (Cycles 4-6), JSTUC Coordinator

JWST+HST Rocky Worlds DDT Core Implementation Team 2025 – Present
Deputy JWST Data Analysis Lead

Observing Experience

James Webb Space Telescope — NIRSpec, NIRCam, MIRI 500+ hours

Hubble Space Telescope — STIS and WFC3 300+ orbits
MMT — SWIRC and ARIES 12 nights
W.M. Keck Observatory — OSIRIS 2 nights
Large Binocular Telescope — LMIRCam 1 night
Morris W. Offit Telescope (JHU) — Optical CCD 1 night
Sommers-Bausch Observatory (CU-Boulder) — Optical CCD 9 nights

Mentorship

Post-Baccalaureate Mentor

- **Seti Norris** — STScI SASP+Intern 2024 – Present
  Co-Author on Lothringer et al. 2024

Undergraduate Research Mentor

- **Ja'Heim Goodwin** — STScI SASP+NAC 2024 – Present
- **Brian Kehoe-Seamons** — Utah Valley University 2022 – Present
  Utah Space Grant Recipient; Co-Author on Lothringer et al. 2024
- **Austin Baldwin** — Utah Valley University 2021 – Present
  UVU URSCA Grant Recipient; Utah Space Grant Recipient
- **Audrey Elison** — Utah Valley University 2023
- **Brayden Roberts** — Utah Valley University 2022 – 2024
  Utah Space Grant Recipient
- **Autumn Winch** — Bryn Mawr College 2020 – 2022
  Senior Thesis; Co-Author on Lothringer et al. 2022; Clinical Research Coordinator, University of Pennsylvania

Dissertation Committee Member

- Natalie Allen, Johns Hopkins University 05/2026
- Arjun Savel, University of Maryland 05/2026
- Rachael Amaro, University of Arizona 08/2024
- Jegug Ih, University of Maryland 06/2024

Teaching (UVU)

Instructor — ASTR-4100: Exoplanets and Brown Dwarfs Fall 2023
Instructor — ASTR-1080: Life in the Universe *(3 sections)* 2022 – 2023
Instructor — PHSC-1000: Survey of Physical Science Spring 2022
Instructor — PHYS-489R: Undergraduate Research *(6 sections)* 2022 – 2023
Instructor — ASTR-1040: Elementary Astronomy *(12 sections)* 2021 – 2023

Other Teaching Experience

Guest Instructor — Exoplanetary Astrophysics, University of Maryland 2024
Guest Instructor — Planets, Life, and the Universe, Johns Hopkins University 2020
Co-Instructor — Exoplanets & Their Atmospheres, Johns Hopkins University 2020
JHU Teaching Academy Certificate 2020
JHU Summer Teaching Institute Workshop 2020
LPL Incoming Graduate Student Mentor 2017 – 2019

Pima Community College GED Prep Math Tutor 2015 – 2016
Graduate Teaching Assistant & Guest Lecturer — PTYS 170B2 Fall 2014

Service and Other Experience

Reviewing:

- Journal Reviewer: AAS Journals, *Astronomy & Astrophysics*, *Nature Astronomy*, *Nature*
- Grant/Proposal Reviewer: Canada Innovation Fund, Hubble Space Telescope, Canadian Time Allocation Committee, NSF Review Panelist, NASA Review Panel Executive Secretary

Leadership and Committees:

- STScI UV Astronomy Workshop Organizer (Spring 2027) 2025 – Present
- Chesapeake Bay Area Exoplanet Meeting Co-Organizer 2024 – Present
- STARGATE Collaboration Executive Committee 2023 – Present
- Atmospheric Escape Observer Pioneers Mission Concept Science Team Lead 2026
- STScI Science Evaluation Committee 2025
- STScI Atmospheric Escape Workshop Co-Organizer 2025
- Aspen CfP JWST Thermal Emission SOC Spring 2025
- STScI-JHU ExoJamboree SOC/LOC Fall 2024
- STScI Summer Program Mentor 2024
- UVU Physics Department Advisory Board 2023 – 2024
- UV-SCOPE MIDEX Mission Concept Science Team 2020 – 2023
- JWST ERS Working Group 2017 – 2024
- JHU/STScI Undergraduate Summer Program Organizer 2020, 2021
- AAS Chambliss Poster Award Judge 2019, 2020
- LPL Men's Diversity and Inclusion Auxiliary 2016 – 2019
- LPL Conference Organizing Committee 2015 – 2017
- Visiting Student, Max Planck Institute for Astronomy 06–07/2016
- Graduate and Professional Student Council Travel Grant Judge 2015

Select Press Coverage & Interviews

The (Unofficial) Exoplanet Dashboard: Space.com

JWST Transiting Exoplanet Community Early Release Science: UVU Press Release; Fox 31 News Good Day Utah; KUER 90.1, Utah NPR

Lothringer & Sing et al. 2022: NASA/Hubble Press Release; Sky & Telescope; The Bad Astronomer, SyFy Wire; The Miami Herald

Benneke et al. 2019: SkyMania; IndiaTV News

In the Community: UVU Wins $1.5M NSF S-STEM Scholarship Program; Hutchings Museum JWST First Images Event; Hutchings Museum JWST Launch Celebration; MAVEN Mission to Mars Launch; No Man's Sky Video Game

Outreach

JWST Subject Matter Expert

Salt Lake County Library Lecture Series (×2); UVU Colloquium (×2); Clarke Planetarium JWST Program; Utah Valley Astronomy Club (×2); Hutchings Museum Institute First Image Event; Hutchings Museum Institute Launch Celebration — 2021 – Present

Latinos of Tomorrow Panelist — 2023

Guest Consultant, *Ranger Rick* Magazine — 2022

STScI Outreach Program

Space Astronomy Summer Program Presenter and Organizer; Easy as Pi — Society of American Military Engineers — 2019 – 2021

Reddit /r/AskScience Panel Member — 2015 – 2018

LPL Outreach Program

Summer Science Saturdays; Tucson Festival of Books; Art of Planetary Science; Bennuval: An Evening of Space, Art, and Music — 2014 – 2019

Exoplanet Lecture Series, Flandrau Planetarium, Tucson, AZ — Apr. 2018

Tucson Amateur Astronomy Association — Mar. 2018

The American International School of Muscat — Science Expert — 2018, 2019

Chaparral High School Career Expert — 2018

The American International School of Muscat, Muscat, Oman — Jan. 2014

LASP MAVEN Launch Outreach — 2014

Professional Affiliations

International Astronomical Union — Since 2025

American Astronomical Society — Since 2014

Phi Beta Kappa Member — Since 2014

Planetary Society Member — Since 2011

Skills

IDL, Python, Fortran, Perl, Bash, Matlab, Mathematica